

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**Department of Computer Science & Engineering****ASSESSMENT TOOL 1 – CONCEPT MAPPING**

Course Code	CSA0314	Course Title	Data Structures
CO Assessed	CO2	Assessment	Assessment Tool 1: Concept Mapping
Weightage	35% of CIA	Total Marks	100
CO4 Statement	<i>Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. (BL2, BL3)</i>		
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Section/Batch	Slot C /	Date	03-10-2026

Code of Conduct:

I, _____, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Ch.sriram

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts	
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established	
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure	
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence	
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand	



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Technical Presentation
CSA0314 - DATA STRUCTURES

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Problem & Requirement

Problem:

- A navigation system connects different cities and locations.
- Each city may have connections with many other cities.
- Roads have different distances or travel times.
- The system needs to find the **shortest route** between two locations.

Requirements:

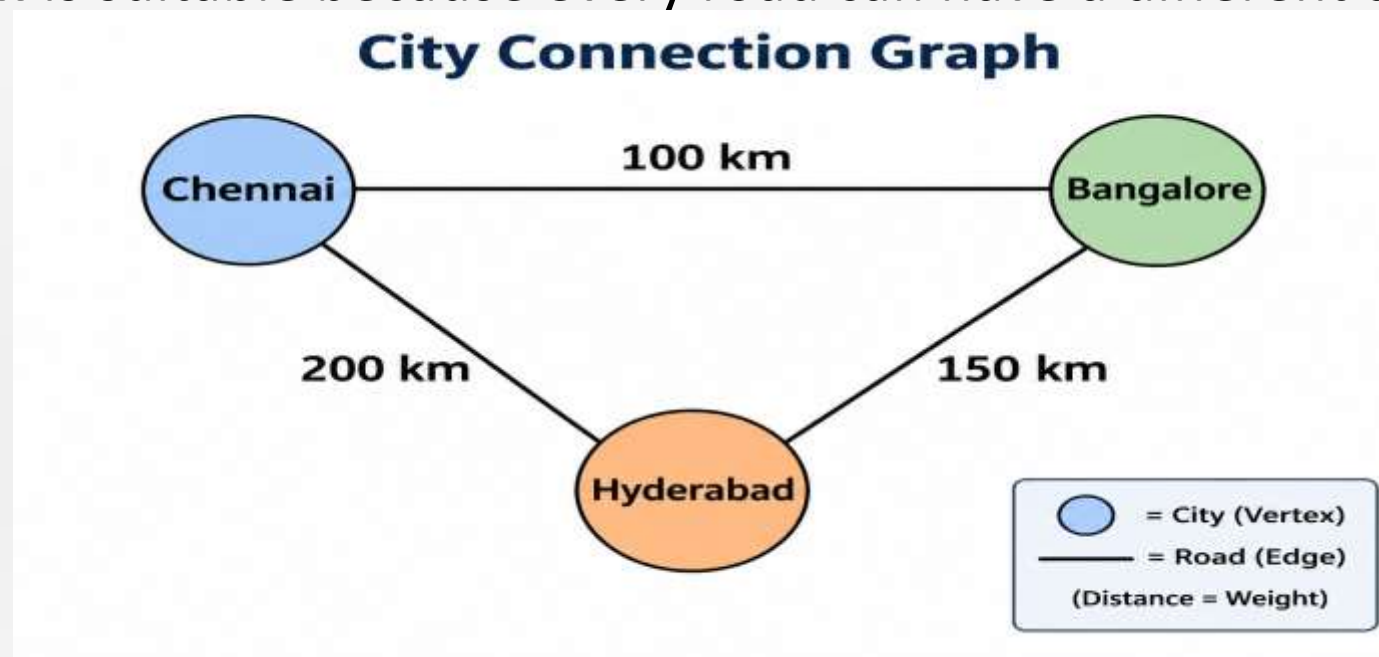
- Store cities.
- Store road connections.
- Store distance/time.
- Find the shortest route.



Suitable Data Structure: Graph

- A **Graph** is a non-linear data structure containing:
- **Vertices (Nodes)** → Cities/locations
- **Edges** → Roads/connections
- **Weights** → Distance or travel time

Weighted Graph is suitable because every road can have a different distance.





Shortest Path Using Dijkstra's Algorithm

Dijkstra's Algorithm can find the shortest route when road weights are non-negative.

Steps:

Select the starting city.

Set its distance as 0.

Set other distances as infinity.

Select the city with the smallest distance.

Update distances of connected cities.

Repeat until the destination is reached.

Example:

$A \rightarrow B = 4 \text{ km}$

$B \rightarrow D = 2 \text{ km}$

Shortest route:

$A \rightarrow B \rightarrow D = 6 \text{ km}$



Why Graph is Appropriate?

Data Structure	Suitability
Array	Less suitable
Linked List	Less suitable
Stack	Not suitable
Queue	Not suitable
Tree	Limited
Graph	Highly suitable



Advantages of Graph

- Represents **city-to-city connections** easily.
- Supports **multiple connections** between locations.
- Stores **distance or travel time** as edge weights.
- Helps find the **shortest route** efficiently.
- Can represent **one-way and two-way roads**.
- Suitable for **large navigation and road networks**.
- Easy to update when **new roads or cities** are added.



Conclusion

- A **weighted graph** is suitable for a navigation system.
- Cities are represented as **vertices**.
- Roads are represented as **edges**.
- Distance or travel time is represented as **weights**.
- **Dijkstra's algorithm** can find the shortest route.
- Graphs are widely useful for GPS, maps, transport and route-planning systems.

